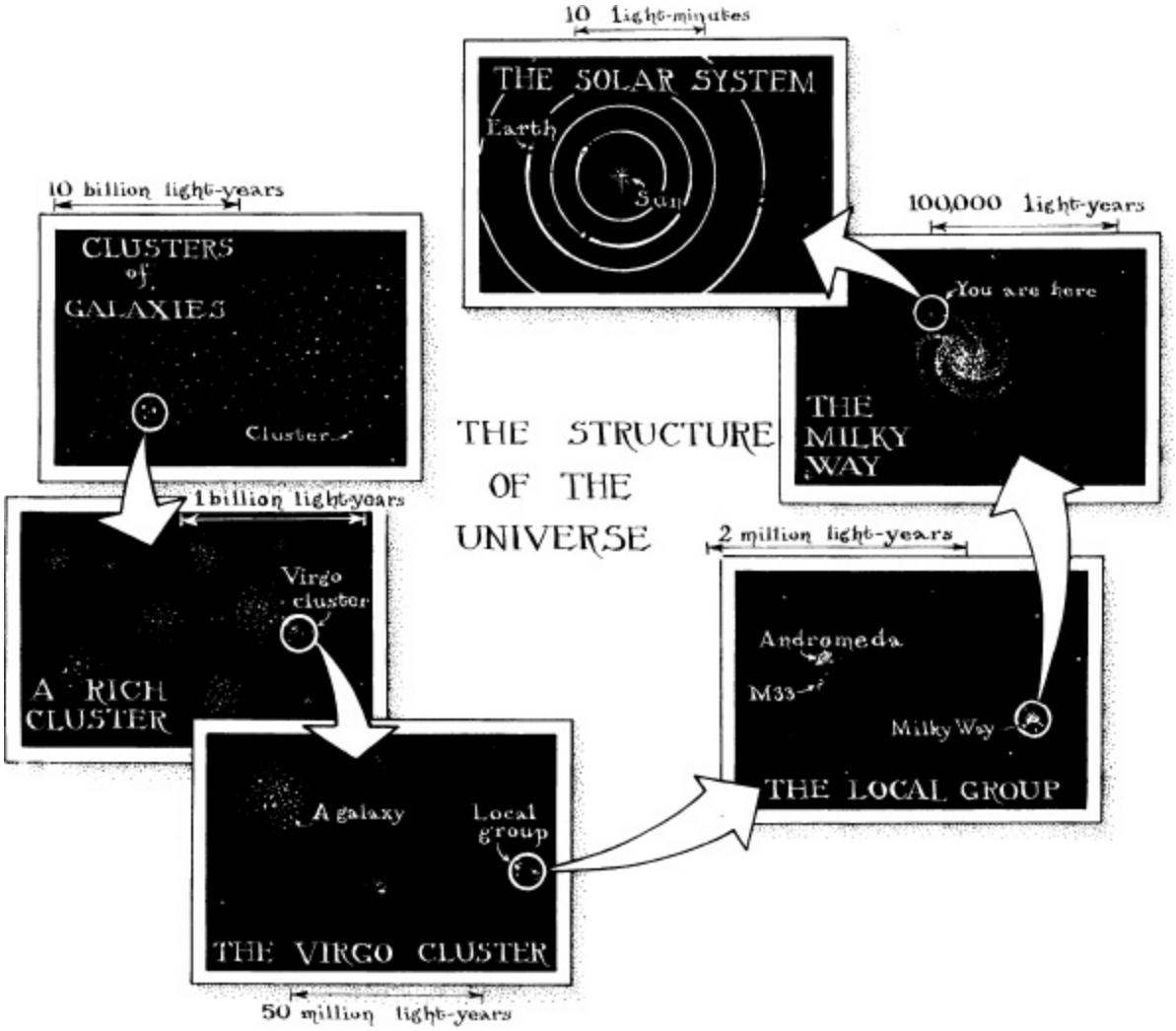


8.1 A sketch of the structure of our Universe.

If searching nearby is so important, then why not search in our own solar system, the realm stretching from the Sun out to the planet Pluto? Might there be a black hole here, among the planets, unnoticed because of its darkness? No, clearly not. The gravitational pull of such a hole would be greater than that of the Sun; it would totally disrupt the orbits of the planets; no such disruption is seen. The nearest hole, therefore, must be far beyond the orbit of Pluto.

How far beyond Pluto? You can make a rough estimate. If black holes are formed by the deaths of massive stars, then the nearest hole is not likely to be much closer than the closest massive star, Sirius, at 8 light-years from Earth; and it almost certainly won’t be closer than the closest of all stars (aside from the Sun), Alpha Centauri, at 4 light-years distance.